

The University of Lahore
Department of Computer Science & IT
Final term Examination



Fall-2024

Course Title:	Data Structures and Algorithms	Course Code:	CS09204	Credit Hours:	4
Course Instructors:	Dr. Ghulam Farooque, Ms. Duaa-e-Nadeem, Ms. Nimra Shafiq, and Ms. Iqra Sabeen	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum Marks:	60		
Date:	18-01-2025	Course Mentor:	Dr. Ghulam Farooque		
Student's Name:		Signature:			
Reg. No:		Section:			

Question # 1: (10 Marks)

Assume you are given a hash table with a size of $m = 12$. To resolve collisions, quadratic probing is used with the following formula:

$$h(k, i) = (h(k) + i^2) \bmod m$$

Where:

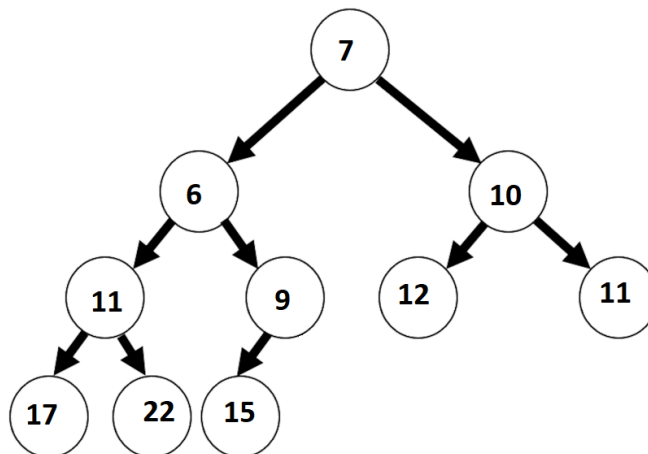
- $h(k) = k \bmod m$ and
- i is the collision attempt index.

Starting with an empty hash table, perform the following steps:

1. Using this approach, insert the keys 15, 3, 19, 37, 32, 24, 43, 48 and 36 step by step into the hash table:
2. For each collision, apply quadratic probing using $F(i) = i^2$ to find the next available position.
3. Show the state of the hash table after each insertion.

Question # 2: (10 Marks)

You are given the tree:



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Tasks:

1. Determine whether the given tree satisfies the properties of a **max heap**; if it does not, construct a valid **max heap**.
2. After constructing the **max heap**, perform the following operations step by step:
 - Insert the value **50** into the **max heap**.
 - Insert the value **2** into the **max heap**.
3. After insertion, perform two deletion operations (removing the root each time) and provide a detailed step-by-step dry run for each deletion operation, showing the heap's state after every step.

Question # 3:

(10 Marks)

You are given the following adjacency matrix:

	1	2	3	4	5	6
1	0	1	0	0	0	0
2	0	0	0	1	0	0
3	0	1	0	0	0	0
4	1	0	1	0	1	0
5	0	1	0	1	0	1
6	1	0	0	0	0	0

Tasks:

1. Construct the directed graph corresponding to the given adjacency matrix.
2. Build the adjacency list representation for the same graph.

Question # 4:

(10 Marks)

You are a software engineer working on a futuristic library management system called "**The Lexicon Vault**". This system ensures the security of book titles by encrypting them. Each book title must be reversed before being stored in the vault as part of the encryption process.

Your task is to design and implement (code) a module that:

1. Accepts a book Title as input.
2. It uses a **stack** (implemented with a **linked list**) to reverse the book title.
3. Outputs the reversed Title.

This approach should be efficient, well-structured, and understandable for future developers.

Example 1	Example 2
Input: Original Title: The Art of Programming	Input: Original Title: Introduction to Algorithms
Output: Reversed Title: gnimmargorP fo trA ehT	Output: Reversed Title: smhtiroglA ot noitcudortnI

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Question # 5: **(10 Marks)**

Starting with an empty tree, construct an **AVL tree** by inserting the values in the given order: **21, 26, 30, 9, 4, 14, 28, 18, 15, 10, and 2.**

For each insertion, perform the following steps:

1. Insert the value into the tree following **binary search tree (BST)** rules.
2. After each insertion, check for any imbalance in the tree. If the tree becomes unbalanced, apply the appropriate **rotation(s)** to restore balance.
3. Provide a **step-by-step dry run** for each operation, showing the state of the tree after every insertion and rotation.

Question # 6: **(10 Marks)**

You are given a binary tree where each node represents a number. Your task is to calculate the sum of all the leaf nodes in the tree. Write a program to perform this task efficiently.

Example 1	Example 2
Input: <pre> 15 / \ 10 20 / \ \ 8 12 25 </pre>	Input: <pre> 15 / \ 10 20 / 8 </pre>
Output: The sum of leaf nodes: 45	Output: The sum of leaf nodes: 28

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GOOD LUCK 😊

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